

Gender Fair 2024 Database Documentation

CSS - <https://ope.ed.gov/campussafety/#/datafile/list>

We use the two following tables from the CSS database:

- Oncampusate202122
- Oncampusvawa202122

Each table had a .sql import script to import its raw data to the MySQL workbench where it can be preprocessed. Both tables share similar columns when it comes to school information (i.e. address, zip, number of people), but they differ in terms of hate crimes and vawa crimes.

The primary key for both tables are "UNITID_P", which is a unique identifier for each institute.

Preprocessing

For either the hate and the vawa table, duplication of records were present as a school can have one OPEID but shares different BRANCH (location). After discussing with the client, it was decided that we would merge the schools that have the same OPEID but different branches.

For instance,

UNITID_P	INSTNM	OPEID	BRANCH	Address	City	State	ZIP	sector_cd	Sector_desc	men_total	women_tot	Total	DOMEST20	DATING20	STALK20	DOMEST21	DATING21	STALK21	DOMEST22	DATING22	STALK22
1.01E+08	Alabama A	100200	Main Cam	4900 MERI	NORMAL	AL	35762	1	Public, 4-y	2369	3638	6007	4	1	0	2	3	1	16	8	1
1.01E+08	University	105200	Main Cam	ADMINIST	BIRMINGH	AL	3.53E+08	1	Public, 4-y	7903	13736	21639	16	20	10	12	13	12	15	45	22
1.01E+08	University	105200	Huntsville	301 Govern	Huntsville	AL	35801	1	Public, 4-y	7903	13736	21639	0	0	0	0	0	0	2	0	0

Note how the 2nd and 3rd records share the same OPEID but different branch location. In the GenderFair2024 DB preprocessing, such records will be merged into one record so that their basic information stays the same (i.e. INSTNM, OPEID, State, etc), and columns such as DATING20 will be added up (in this case 20 + 0) and the total value will be used as the final value in the combined column. In this way, we can remove duplicates without losing data and account for every crime recorded.

We also combined all columns that represent the number of crimes into one column as the total number of crimes for an institute. For instance using the same image, we would summing values from "DOMEST22" to "STALK22" so that we get the total number of crimes committed in 2022. We would then use this number as the yearly number of crimes committed by an institute. This data will then be used to assess the safety score of an institute.

Note that this process would be different for hate and vawa. Hate has about 378 columns dedicated towards different categories of hate crimes committed by different types of people, and there are certain columns that represent the total amount of a crime category. For instance, MURD22 is the sum of MURD_RAC22, MURD_REL22, etc. Therefore, it is important to not over-count the number of crimes by adding MURD20 with its child columns, which would result in doubling the number of hate crimes in total. For vawa, all relevant columns are independent, therefore a simple addition would produce the total crimes

Finally, we want to add a column that showcases the average crimes committed per 1000 students for an institute per request from our client. In order to do so, we use the average number of crimes committed per year and divide by the “Total” column, then multiply the result by 1000. This effectively gives us the average crimes committed per 1000 students. After the above steps the two CSS tables should consist of unduplicated records that can use OPEID as a unique identifier. The old tables will either be updated to use OPEID as the new PK, or new tables will be created with OPEID as the PK. This can help the merging process later with the IPEDS database. The two CSS tables should now be ready to merge.

IPEDS - <https://nces.ed.gov/ipeds/use-the-data/download-access-database>

The IPEDS database consists of many tables that have information about most US post-secondary institutes. The tables that will be used in making the IPEDS table consist of:

- S2022_OC
- C2022_B
- EF2022
- HD2022
- S2022_IS
- IC2022
- SAL2022_IS

The description of each table can be found in the IPEDS202223TablesDoc downloaded from the data source.

For the tables selected, all of them contain information that either directly relates to our project or information that helped us filtering data. The primary key for all tables are UNITID.

Preprocessing

S2022_IS

This table contains full-time instructional staff, by faculty and tenure status, academic rank, race/ethnicity, and gender (Degree-granting institutions): Fall 2022. Preprocessing of this table includes filtering the table using the “SISCAT” column to obtain the type of staff we are interested in (Professors, Associate profs, Assistant profs, etc), as well as the total number of staff of an institute.

S2022_OC

This table contains full- and part-time staff by occupational category, race/ethnicity, and gender: Fall 2022. Preprocessing of this table includes filtering the table using the “OCCUPCAT” column to obtain the number of non-teaching staff with various race/ethnicity, grouping all kinds of non-teaching staff on UNITID to ensure the uniqueness of an institute. The rationale of the choice in OCCUPCAT (230, 250, 300, 310, 320, 330, 340, 350, 360, 370, 380, 390) can be found in IPEDS202223TablesDoc.xlsx via variable descriptions.

SAL2022_IS

This table contains number and salary outlays for full-time non medical instructional staff, by gender, and academic rank. Preprocessing of this table includes filtering the table on column "ARANK" to obtain salary outlays for instructional staff and computing the average salary outlay for men and women within an institute. This is achieved by dividing the "SAOUTLM(W)" column with "SAINSTM(W)" column. Note that in some cases the average salary for M/W can be 0 due to "SAINSTM"/"SAINSTW" having a value of 0. In these cases, we simply assign the average salary to NULL for better accuracy.

IC2022

This table contains educational offerings, organization, admissions, services and athletic associations. Preprocessing of this table includes filtering the table using the "LEVEL" columns to obtain institutes that offer degrees above or equal to an associate's degree, filtering the table using the "CNTLAFFI" column to obtain public and private non-profit institutes (1 for public, 3 for private religious affiliated, 4 for private non-religious affiliated), and selecting "STUSRV8" column that indicates whether a school offers daycare or not.

C2022_B

This table contains Number of students receiving awards/degrees, by race/ethnicity and gender. Preprocessing of this table includes selecting the column "CSTOTLT" to obtain the number of people who received a degree from an institute.

EF2022

This table contains Gender, attendance status, and level of student: Fall 2022. Preprocessing of this table includes filtering the table with column "EFLEVEL" to select all student types, selecting "EFTOTAL" to obtain the total number of students in an institute.

HD2022

This table contains Directory information of all institutes. This table is not preprocessed but used in merging all IPEDS table to provide basic information of a school (i.e. name, etc)

Merging Tables

The merge process of all relevant tables mentioned above.

First, we select the schools that we actually want to have in our IPEDS table. The standard of assessing such schools are:

- The school has to have at least 200 students (satisfied by setting EFTOTAL >= 200 in EF2022)
- The school has to be public/private non-profit (satisfied by choosing schools that has the value 1,3, and 4 in "CNTLAFFI" column in IC2022)
- The school has to offer degrees higher or equal to an associate's degree (satisfied by choosing schools that offers "LEVEL3,4,5,6,7,8,17,18, and 19" in IC2022)

- The school has to have at least 50 graduates per year (satisfied by setting CSTOTLT >= 50 in C2022_B)

Joining the EF2022, IC2022, and C2022_B with an inner join would give us the list of schools' UNITID. From here, another inner join is performed with HD2023 to give the IPEDS table directory information of an institute.

Subsequently continued the merging process with other tables (SAL2022_IS, S2022_OC, S2022_IS) via inner joins and obtained a fully merged IPEDS table. Select only the variable that we plan to display on the front-end using a metadata documentation.

* When joining with S2022_IS, we will see that the data are stored in this way:

	UNITID	SISCAT	HRTOTLT	HRTOTLM	HRTOTLW	PRFPERCW	BKAAPERCW	HISPERCW	AISAPERCW
0	100654	101	40	31	9	0.225000	0.125000	0.000000	0.025000
1	100654	102	53	31	22	0.415094	0.245283	0.000000	0.113208
2	100654	200	73	53	20	0.273973	0.178082	0.000000	0.068493

Note that the same UNITID (institute) has multiple records with different SISCAT values, meaning that it uses both UNITID and SISCAT as PK. However, in our database setting we prefer to let the same UNITID (institute) to have one record, like this:

	UNITID	PRFPERCW_101	PRFPERCW_102	PRFPERCW_200	BKAAPERCW_101	BKAAPERCW_102	BKAAPERCW_200	HISPERCW_101	HISPERCW_102	HISPERCW_200	AISAPERCW_101	AISAPERCW_102	AISAPERCW_200
0	100654	0.225000	0.415094	0.273973	0.125000	0.245283	0.178082	0.000000	0.000000	0.000000	0.025000	0.113208	0.068493

Which effectively augmented the SISCAT column into 3 different columns (PRFPERCW_101, PRFPERCW_102, PRFPERCW_200, etc) to reflect its distinction (between 101, 102, and 200). This allows us to use only UNITID as the PK and therefore maintain our database structure.

If we join the table directly, there will be multiple records for each UNITID (institutes), therefore not able to produce a single UNITID PK within the school. Therefore a flattening of the column SISCAT in S2022_IS should be performed to convert/pivot the data table in a correct way so that the join operation can be performed in the desired way.

After this, the IPEDS table should be ready to merge.

IRS

All relevant IRS code should be accessed through this link: hit-shirakrk/irs-990-parser

Merge All Tables

To merge the two CSS table with the IPEDS table and the IRS table, we need to do it in two steps:

- Merge IPEDS with CSS using OPEID

- Merge IPEDS+CSS with IRS using EIN

Merging IPEDS with CSS using OPEID is straightforward, which is an inner join on OPEID. This should not produce any duplicate data.

Merging IPEDS+CSS with IRS using EIN involves a left join so that we can keep all data from IPEDS+CSS but also include data from the IRS table if it exists. If a school has no data in the IRS table, all of the IRS columns of that school will be NULL. However, since the IRS data include multiple records from a single EIN, we need to first deduplicate the IRS table before joining with the IPEDS+CSS to avoid PK error.

The deduplication process involves choosing which IRS_MONTH we want to keep as some schools file reports for different IRS_MONTH. The current method is to group records by EIN and check through each IRS column. If the column has no data in any reports by any IRS_MONTH, then leave it as empty. If the column has multiple values throughout all reports filed in all IRS_MONTH, use the latest value. In this way we ensure that each school has the most up-to-date IRS data to display as well as maximize the possibility for a school to have IRS data.

After deduplication of the IRS table, the merging process should be performed with no further issues, and a usable table should be populated in the workbench. Any change beyond this point should be done by prospect developers.